

$$\begin{aligned}
& \frac{1+z}{1-z} \\
&= \frac{1+x+yi}{1-x-yi} \\
&= \frac{1+x+yi}{(1-x)^2+y^2} \times (1-x+yi) \\
&= \frac{(1+x+yi)(1-x+yi)}{(1-x)^2+y^2} \\
&= \frac{1-x+yi+x-x^2+xyi+yi-xyi-y^2}{(1-x)^2+y^2} \\
&= \frac{1-x^2-y^2+2yi}{(1-x)^2+y^2}
\end{aligned}$$

using the given restriction on the absolute value: